MARYLAND INSTITUTE OF TECHNOLOGY AND MANAGEMENT

(CHORINDA, GALUDIH, JAMSHEDPUR, JHARKHAND)

DEPARTMENT OF MECHANICAL ENGINEERING

Basic Mechanical Engineering [PCME2]

EVEN SEMESTER

MODULE - 1

STUDY NOTES

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Metallic and non-metallic properties

Mechanical properties

*Strength: – In mechanical engineering, "strength" refers to a material's ability to resist applied forces without failing whether by yielding, fracturing, or deforming permanently. It is a fundamental property that determines how materials and structures perform under different loading conditions such as tension, compression, shear, torsion, and bending. For metals that experience gradual elastic-plastic transition, the point of yielding may be determined as the initial departure from linearity of the stress-strain curve; this is sometimes called the proportional limit, The stress corresponding to the intersection of this line and the stress-strain curve as it bends over in the plastic region is defined as the yield strength, After yielding, the stress necessary to continue plastic deformation in metals increases to a maximum, and then decreases to the eventual fracture. The tensile strength TS (MPa or psi) is the stress at the maximum on the engineering stress-strain curve. This corresponds to the maximum stress that can be sustained by a structure in tension; if this stress is applied and maintained, fracture will result. All deformation up to this point is uniform

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throughout the narrow region of the tensile specimen. However, at this maximum stress, a small constriction or neck begins to form at some point, and all subsequent deformation is confined at this neck, this phenomenon is termed "necking," and fracture ultimately occurs at the neck. The fracture strength corresponds to the stress at fracture.

*Toughness: – Toughness is a mechanical term that is used in several contexts; loosely speaking, it is a measure of the ability of a material to absorb energy up to fracture. Specimen geometry as well as the manner of load application are important in toughness determinations. For dynamic (high strain rate) loading conditions and when a notch (or point of stress concentration) is present, notch toughness is assessed by using an impact test. Furthermore, fracture toughness is a property indicative of a material's resistance to fracture when a crack is present. For the static (low strain rate) situation, toughness may be ascertained from the results of a tensile stress–strain test. It is the area under the–curve up to the point of fracture. The units for toughness are the same as for resilience (i.e., energy per unit volume of material). For a material to be tough, it must display both strength and ductility; and often, ductile materials are tougher than brittle ones. Hence, even though the brittle material has higher

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yield and tensile strengths, it has a lower toughness than the ductile one.

*Brittleness: – Brittleness is the property of a material that causes it to fracture suddenly with little or no plastic deformation when stress is applied. In simple terms, brittle materials break easily rather than bending or stretching. It is essentially the opposite of ductility. While ductile materials can stretch or deform before breaking, brittle materials fail abruptly.

*Creep: – Creep is the slow, time-dependent deformation of a solid material under constant stress, often occurring at high temperatures and below the material's yield strength. It is critical in engineering because it can cause gradual failure in components exposed to long-term loads. Unlike sudden fracture, creep is progressive and time-dependent, making it especially dangerous in long-term applications.

*Fatigue: – Fatigue in materials science refers to the progressive weakening and eventual failure of a material due to repeated cyclic loading, even when the stress levels are well below its ultimate tensile strength. It is one of the most common causes of engineering failures. Unlike creep

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(time-dependent deformation) or brittleness (sudden fracture), fatigue occurs gradually through repeated stress cycles.

*Stiffness: – Stiffness is a measure of how resistant a material or structure is to deformation when subjected to an applied force. Stiffness tells us how much a material resists being deformed, not how much load it can ultimately carry.

*Ductility: – Ductility is the ability of a material to undergo significant plastic deformation before fracture, usually measured by how much it can be stretched into a wire or elongated under tensile stress. It is essentially the opposite of brittleness.

*Malleability: – Malleability is the ability of a material to be hammered, rolled, or pressed into thin sheets without breaking or cracking. It is a measure of how well a material can withstand compressive forces while deforming plastically. Closely related to ductility, but while ductility refers to stretching into wires, malleability refers to flattening into sheets.

*Elasticity: – Elasticity is the property of a material that enables it to return to its original shape and size after the removal of an applied force, provided the deformation was within its elastic limit. Governed by Hooke's Law:

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$$\sigma = E \cdot \varepsilon$$

Where, σ = stress, E= Young's Modulus and ε = strain,

*Plasticity: – Plasticity is the property of a material that allows it to undergo permanent deformation without breaking when subjected to external forces. Unlike elasticity, where a material returns to its original shape after the load is removed, plasticity means the material retains its new shape. It is the foundation of processes like forging, rolling, extrusion, and moulding.

Physical properties

*Density: – Density is the physical property of matter that expresses how much mass is contained in a given volume. It tells us how "heavy" or "light" a material feels for its size. Units: Commonly expressed in kg/m^3 or g/cm^3

*Viscosity: – Viscosity is the property of a fluid that describes its resistance to flow. In simple terms, it tells us how "thick" or "sticky" a liquid is. Temperature dependent: Liquids generally become less viscous when heated; gases become more viscous when heated.

*Colour: – Colour is the property of a material or object that describes the way it interacts with light, as perceived by the

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human eye. It is essentially the visual sensation produced when light of certain wavelengths is reflected, absorbed, or transmitted. Human eyes detect colour using cone cells sensitive to red, green, and blue light.

*Finish: – Finish refers to the surface quality or appearance of a material after it has been processed or manufactured. It describes how smooth, shiny, textured, or polished the surface is, and it plays a big role in both aesthetics and functionality. It can be functional (improving durability, corrosion resistance, or friction) or decorative (enhancing appearance).

*Porosity: – Porosity is the property of a material that describes the presence of tiny pores, voids, or spaces within its structure. It indicates how much empty space exists inside a solid and is usually expressed as a percentage of the total volume. Porous materials allow fluids or gases to pass through.

*Specific gravity: – Specific gravity is the ratio of the density of a substance to the density of a reference substance, usually water (for liquids and solids) or air (for gases). It is a dimensionless quantity, meaning it has no units. For liquids and solids, the reference is typically water at 4°C (density = 1 g/cm^3). For gases, the reference is usually air.

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*Fusibility: – Fusibility is the property of a material that describes how easily it can be melted when heat is applied. It indicates the readiness of a substance to fuse or liquefy under high temperature. It is often measured by the melting point, lower melting point means higher fusibility. Alloys often have different fusibility compared to pure metals.

Thermal properties

*Specific heat: – Specific heat is the amount of heat energy required to raise the temperature of a unit mass of a substance by one degree Celsius (or one Kelvin). It tells us how much energy a material can store as heat. Units: J/(kg·K).

*Thermal conductivity: – Thermal conductivity is the property of a material that describes how efficiently it can transfer heat. It tells us how quickly heat flows through a substance when there is a temperature difference. Units: W/(m·K).

*Thermal resistance: – Thermal resistance is the property of a material or system that describes how strongly it resists the flow of heat. It is essentially the "opposite" of thermal conductivity: while conductivity measures how easily heat passes through, resistance measures how much the material slows it down. Units: K/W (Kelvin per Watt).

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*Thermal diffusivity: – Thermal diffusivity is the property of a material that describes how quickly heat spreads through it relative to its ability to store heat. It combines both thermal conductivity and specific heat capacity into one measure. Units: m^2/s.

Magnetic properties: – Magnetic properties describe how materials respond to a magnetic field and whether they can be magnetized. These properties depend on the arrangement of electrons and atomic structure.

Electrical Properties: – Electrical properties describe how materials respond to electric fields, currents, and charges. They determine whether a material conducts electricity, resists it, stores it, or interacts with it in other ways.

*Resistance and Resistivity: – Resistance and resistivity are closely related electrical properties, but they describe different aspects of how materials oppose the flow of electric current. Resistance is the opposition offered by a specific object or conductor to the flow of electric current. Units: Ohms (**Ω**). Depends on: Material, length, and thickness of the conductor. Resistivity is the intrinsic property of a material that quantifies how strongly it resists current flow. Units: Ohm-meter (**Ω**·m).

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*Conductance and Conductivity: – Conductance and conductivity are related electrical concepts, but they describe different levels of behaviour—one applies to a specific object, the other to the material itself. Conductance is the measure of how easily a particular object or component allows electric current to flow. Units: Siemens (S). Conductivity is the intrinsic property of a material that quantifies its ability to conduct electricity. Units: Siemens per meter (S/m).

*Capacitance: – Capacitance is the property of a system that describes its ability to store electric charge when a voltage is applied. It's a fundamental concept in electronics, especially in capacitors.

*Chemical properties: – In mechanical engineering, chemical properties of materials refer to their ability to interact with the environment and other substances, which directly affects their performance and longevity. These properties include corrosion resistance, oxidation resistance, chemical stability, and reactivity with acids, alkalis, or solvents. For instance, stainless steel resists corrosion due to the presence of chromium, while mild steel is more prone to rusting. Similarly, ceramics are chemically stable and inert, making them suitable for high-temperature applications, whereas

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polymers may degrade when exposed to certain chemicals or ultraviolet radiation. Alloying elements such as nickel, molybdenum, and chromium are often added to metals to enhance their chemical behaviour, improving resistance to acids or oxidation. These chemical properties form the foundation for further classification of materials, as they determine suitability for specific environments like marine, industrial, or high-temperature conditions, and guide engineers in selecting the right material for reliable and safe applications

*Corrosion resistance: – Corrosion resistance is one of the most important chemical properties of engineering materials, as it defines their ability to withstand deterioration when exposed to moisture, oxygen, salts, acids, or other aggressive environments. Materials with high corrosion resistance, such as stainless steel, aluminium, and certain alloys, form protective surface films that prevent further chemical attack, ensuring long-term durability. On the other hand, materials like mild steel or iron are more prone to rusting and require protective coatings, paints, or galvanization to extend their service life. The level of corrosion resistance directly influences where a material can be used for example, stainless steel is preferred in food processing, medical tools,

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and marine applications, while aluminium is widely used in aerospace and automotive industries due to its resistance to oxidation. Engineers must carefully evaluate corrosion resistance when selecting materials, as failure to do so can lead to structural damage, safety hazards, and costly maintenance. This property also serves as a basis for further classification of materials, distinguishing those suitable for harsh chemical environments from those that need protective measures.

*Acidity and alkalinity: – Acidity and alkalinity as chemical properties describe how materials behave when exposed to environments with different pH levels, which is a measure of how acidic or basic a solution is. Materials that come into contact with acidic environments may undergo chemical attack, leading to weakening or degradation; for example, concrete is vulnerable to acidic solutions but remains stable in alkaline conditions. On the other hand, metals like aluminium are stable in neutral environments but can react strongly with alkaline solutions, while stainless steel resists many acidic and alkaline attacks due to its protective oxide layer. Understanding how materials respond to acidity and alkalinity is crucial in mechanical engineering, especially for applications in chemical plants, marine

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structures, and construction, where exposure to varying pH levels is common. This property helps engineers classify materials based on their suitability for specific environments, ensuring that the chosen material can withstand chemical exposure without losing strength or functionality.

*Ferrous and non-ferrous metals: - Ferrous and non-ferrous metals are two broad classifications of engineering materials based on the presence or absence of iron. Ferrous metals contain iron as their main constituent and are generally strong, durable, and magnetic, making them widely used in construction, automotive, and machinery. Examples include mild steel, cast iron, and stainless steel. However, they are often prone to corrosion unless alloyed with elements like chromium or nickel to improve resistance. On the other hand, non-ferrous metals do not contain iron and are usually lighter, more resistant to corrosion, and non-magnetic. Common examples are aluminium, copper, brass, and titanium, which are used in aerospace, electrical wiring, and decorative applications. The distinction between ferrous and non-ferrous metals is important in mechanical engineering because it helps in selecting materials based on strength, weight, corrosion resistance, and specific functional requirements. This classification also guides further study into their

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mechanical and chemical properties, ensuring that engineers choose the right material for each environment and application.

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